LNMIIT
The LNM Institute of Information Technology

The LNM Institute of Information Technology, Jaipur
CC221: Computer Communication Networks
2024-25 Even Semester
End-Term Exam

Max Marks: 80 **Date: 5th May 2025** **Time: 180 Mins.**

Total Questions	Total Marks	CO1	CO2	CO3	CO4	CO5
12	80	Q.1	Q.2, Q.3	Q.4, Q.5, Q.6, Q.7	Q.8, Q.9, Q.10, Q.11	Q.12
CO weightage		6/80=7.5%	10/80=12.5%	26/80=32.5%	28/80=35%	10/80=12.5%

Instructions:

a. *There are total 12 questions, and all the questions are mandatory. You can use calculators.*
b. *Attempt questions in the given order. Make and state assumptions precisely, as and when required. All the notations have their usual meanings.*

Q.1) A guy named Tom explains that encoding is the process of using various patterns of voltage or current levels to represent the 1s and 0s of a digital signal on a transmission link. Tom instructs Jerry to use Bipolar RZ, Manchester encoding, and Differential Manchester encoding (where 0 results in a transition and 1 result in no transition) to encode the data stream **110101**. As Jerry, your task is to complete the encoding using the specified schemes. **[2+2+2]**

Q.2) The three 16-bit words at the sender are given as **0110011001100000, 0101010101010101,** and **1000111100001100**. What is the actual data stream that the sender would transmit after calculating the checksum? How will the receiver verify whether the data stream obtained is correct or not? **[2+3]**

Q.3) What is the capture effect in CSMA/CD? Following CSMA/CD, Station A and Station B fight for the channel, and the first two backoffs are won by Station B. Determine the probability of Station A winning the third backoff. **[2+3]**

Q.4) A datagram of 4000 bytes, including a 40-byte header, arrives at a router. The router must forward the datagram over a link with a Maximum Transmission Unit (MTU) of 580 bytes. How many fragments must the router create? Determine the total length of each fragment, and the fragmentation offset? **[1+2+3]**

Q.5.a) Consider the network with link costs shown in **Fig.1**. Use Distance Vector routing protocol to find the minimum cost from each node to the destination node **4** and the next node along the shortest path. Use label (n, D_i) for each node i, where n is the next node along the shortest path, and D_i is the current minimum cost from node i to the destination. Show the result in an appropriate form. **[2]**

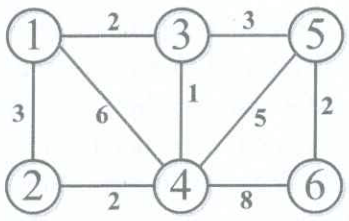


Fig. 1

(b) Let's assume the minimum cost to destination node **4** is determined from all network nodes, but now the link between **3** and **4** goes down. Determine the new route from each network node to destination node **4** step by step in an appropriate form followed by the last step you got in part (a). **[4]**

Q.6) A network with four routers with link costs is shown in **Fig.2**. Determine the distance vector at each router after network is in stable state. Now, let's assume that the link between '**x**' and '**y**' is broken, how will the distance vector of all the routers modify? Will this persist in any problem? How could that problem be resolved? Give detail answer step by step? **[6]**



Fig. 2

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Q.7.a) Consider the network with link costs shown in **Fig.3**. Using Dijkstra's shortest path algorithm, find the shortest path from source node 'F' to all other nodes step by step in an appropriate form. **[6]**

(b) Taking node 'F' as the root, draw the shortest-path tree from node 'F' to all network nodes using the results obtained in part (a)? **[2]**

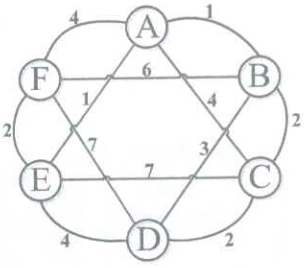


Fig. 3

Q.8) Using example of sending the segments from a client to the server, define the role of SYN, PSH, URG, FIN, and RST bits present in the TCP header. **[5]**

Q.9) a) In a general queueing model where packets arrive independently, prove that the arrival process follows Poisson distribution.

b) The timeline is divided into 10 intervals, and in each interval, the probability of a packet arrival is 0.5. What is the probability that a total of 4 packets will arrive? **[5+3]**

Q.10) Consider a TCP connection between a client and a server having the round-trip time (RTT) is 10 ms. This RTT applies to the last segment sent, considering the sender's window size. The size of the receiver's advertised window is 200 KB. The slow-start threshold at the client is 162 KB. The maximum segment size is 5 KB. The connection is established at time t = 0. Assume there are no timeouts or errors during transmission. Show the proper timeline diagram of the segment's transmission from client to the server till the threshold is reached. What is the size of the congestion window (in KB) at time (t+50) ms, and (t+100) ms, respectively after all the acknowledgments have been processed?

[5+3]

Q.11) While handling the TCP congestion, the receiver window size advertised is 128 segments and the slow-start threshold is set to 70 segments. The slow-start phase starts (always) with 4 segments and grows exponentially till the threshold is reached. After this, the additive increase procedure starts and the three ACKs event happens when the sender window size reaches 70 segments. Followed by this, the time-out occurs when the sender window size is 38 segments. Going further, the data transfer process stops when the sender window size reaches 20 segments. To illustrate this process, draw the diagram with the number of rounds on the ***x-axis*** and the congestion window size (cwnd) on the ***y-axis***, showing how TCP congestion is managed, and explain how you made the diagram with proper justification. **[7]**

Q.12) a) With the help of non-persistent HTTP, you need to download 10 objects from the URL ***www.lnmiit.in/department/cce/projects***. Explain the steps involved in the process. **[5]**

b) What is cellular network and its functionality? **[5]**